
UNIFIED MECHANICS

Volume II

Carrier Geometry, Fields, and Gravitation

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ABSTRACT

This volume develops the finite carrier, its exceptional geometry, admissible reductions, readable algebras, field transport, and the proposed bridge to gauge and gravitational dynamics. Established lattice and Lie theory are kept distinct from the framework's physical selection claims.

Contents

1	THE CARRIER PROBLEM	1
1.1	From abstract traversal to a concrete representation	1
1.2	Why the inherited evenness argument is incomplete	2
2	THE E_8 ROOT CARRIER	3
2.1	Coordinates and primitive crossings	3
2.2	Triangular closure	3
2.3	Complete finite closures	4
2.4	The Coxeter plane and golden structure	4
3	FINITE SPHERICAL DESIGNS	5
3.1	The 24-cell and 600-cell	5
4	ADMISSIBLE REDUCTION AND THE READABLE ALGEBRA	6
4.1	The operator-algebraic construction	6
4.2	Minimal and graded decay	7
4.3	The rank-three candidates	7
4.4	The $A_3 \times D_5$ branching	8
5	CONNECTIONS, HOLONOMY, AND THE CO-DISTINCTION PROPOSAL	9
5.1	Established geometric structure	9
5.2	The Shields field as a proposed extension	9
6	THE GRAVITATIONAL SQUARE	11
6.1	MacDowell–Mansouri gravity	11
6.2	The corpus coefficient proposal	11
6.3	Variation and torsion	12
7	LEAST BOUNDARY AND THE COMPRESSION FIELD	13
7.1	The single-cell theorem	13
7.2	From isolated cells to saturation	14
7.3	The Bubble Field as a compression map	14
7.4	Spherical metric reconstruction	15

8	THE SOURCE SECTOR AND PRESENT FRONTIER	16
8.1	Local and bilocal fields	16
8.2	What is exact, what is proposed	16
8.3	Forward obligations	17

The carrier problem

1.1 From abstract traversal to a concrete representation

Volume I defined a traversal without specifying the set on which it acts. A concrete carrier must add at least four ingredients:

1. a state or displacement space;
2. a composition law for admissible steps;
3. an inner product, norm, or other invariant that distinguishes primitive from composite steps;
4. a symmetry group under which equivalent descriptions agree.

The corpus proposes a finite, self-dual carrier and identifies its primitive crossings with the shortest nonzero vectors of a lattice. That proposal is mathematically productive, but the logical order matters. Self-duality alone does not imply that the carrier is a positive-definite integral lattice; integrality does not imply evenness; and none of these properties fixes the rank. They are separate premises or intermediate obligations.

DEFINITION 1.1 — LATTICE CARRIER

A lattice carrier is a full-rank discrete subgroup $L \subset \mathbb{R}^n$ equipped with the Euclidean inner product. Its dual is

$$L^* = \{x \in \mathbb{R}^n : \langle x, y \rangle \in \mathbb{Z} \text{ for every } y \in L\}. \quad (1.1)$$

It is *unimodular* when $L = L^*$, and *even* when $\langle x, x \rangle \in 2\mathbb{Z}$ for all $x \in L$.

Positive-definite even unimodular lattices exist only in dimensions divisible by eight. In dimension eight there is, up to isometry, one such lattice: E_8 (Conway and Sloane, 1999). This is a strong classification theorem. It becomes a carrier-selection theorem only after the framework derives or stipulates positive definiteness, integrality, evenness, unimodularity, and minimal dimension eight.

 CONDITIONAL SELECTION 1.1 — THE FIRST EVEN UNIMODULAR CARRIER

If the carrier is required to be a positive-definite even unimodular lattice, and if the realised rank is the smallest positive admissible rank, then it is isometric to the E_8 lattice. The mathematical uniqueness is established; the physical premises remain framework commitments.

1.2 Why the inherited evenness argument is incomplete

The corpus motivates even norm by saying that a relation read outward and back contributes twice to its self-pairing. This can become a proof only after the pairing is defined and shown to take integer values on all carrier vectors. Without integrality, “twice” gives a real number, not an even integer. Without a construction showing that every lattice vector is assembled from such two-sided relations, the argument covers only selected generators.

Accordingly, this edition treats evenness as an open representation lemma:

 RESEARCH TARGET 1.1 — DERIVE THE LATTICE HYPOTHESES

Construct a functor or representation from the traversal grammar to a lattice L and prove, rather than name, that (i) the pairing is integral, (ii) all norms are even, (iii) $L = L^*$, (iv) the form is positive definite, and (v) the minimal faithful rank is eight. Only then would the E_8 selection be derived from the foundation.

The E_8 root carrier

2.1 Coordinates and primitive crossings

In a standard normalization the E_8 roots have squared norm two. They consist of two families:

$$\Phi_{\text{mint}} = \{(\pm 1, \pm 1, 0, \dots, 0) : \text{all permutations}\}, \quad (2.1)$$

$$\Phi_{\text{half}} = \left\{ \frac{1}{2}(\pm 1, \dots, \pm 1) : \text{an even number of minus signs} \right\}. \quad (2.2)$$

There are 112 roots in the first family and 128 in the second, for a total of 240. Their integral span is the E_8 root lattice. The root system is invariant under the Weyl group generated by reflections

$$s_\alpha(x) = x - 2 \frac{\langle x, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha. \quad (2.3)$$

The corpus identifies these roots with primitive crossings. Mathematically, they are the shortest nonzero lattice vectors. Physically, the identification requires an action or transition rule showing why a shortest lattice vector is an indivisible event rather than merely a convenient coordinate step.

2.2 Triangular closure

If roots $\alpha, \beta \in \Phi(E_8)$ satisfy $\langle \alpha, \beta \rangle = -1$, then $\alpha + \beta$ is also a root. Setting $\gamma = -(\alpha + \beta)$ gives

$$\alpha + \beta + \gamma = 0. \quad (2.4)$$

The parallelogram spanned by α and β has Gram determinant

$$\det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3, \quad (2.5)$$

so its area is $\sqrt{3}$, while the triangle bounded by the three roots has area $\sqrt{3}/2$. Earlier corpus language sometimes called both the root cell. The distinction is fixed here.

INTERNAL RESULT 2.1 — MINIMAL NONDEGENERATE ROOT POLYGON

Within an admissible-step model in which roots are steps and immediate reversal is excluded as degenerate, a three-root relation $\alpha + \beta + \gamma = 0$ is the shortest possible closed polygon. Its geometric existence is exact. Its interpretation as the minimal history of physical closure is conditional on the step model.

If one attaches the proposed retained factor r to each step, the three-step product is the exact number $r^3 = (\sqrt{5} - 2)/8$. This is reuse of the foundational prior, not an independent recovery.

2.3 Complete finite closures

Because every root occurs with its negative,

$$\sum_{\alpha \in \Phi(E_8)} \alpha = 0. \quad (2.6)$$

Thus a sequence using every root exactly once has zero net displacement. The corpus also supplies a computational witness satisfying its additional local sequencing rule. Volume V records the witness, code, hash, and verifier. Existence of one such ordering does not establish uniqueness, count all admissible closures, or show that nature executes one.

2.4 The Coxeter plane and golden structure

The E_8 Coxeter element has order thirty. Projection of the root system onto an associated Coxeter plane places the 240 roots on eight rings of thirty. In the corpus's numerical certificate, the ring radii pair into four ratios equal to φ within floating-point tolerance. This geometry belongs to established relationships among Coxeter systems, quasicrystalline projections, and the noncrystallographic group H_4 (Elser and Sloane, 1987; Koca et al., 1998; Coxeter, 1973).

ESTABLISHED CONTACT 2.1 — E_8 , H_4 , AND THE GOLDEN FIELD

The E_8 lattice admits four-dimensional cut-and-project descriptions with H_4 symmetry, and quaternionic formulations decompose its roots into golden-related H_4 sets (Elser and Sloane, 1987; Koca et al., 1998). The golden ratio is therefore genuine carrier geometry; its presence does not establish a physical vacuum or coupling.

Finite spherical designs

3.1 The 24-cell and 600-cell

The 24-cell is the regular four-polytope associated with the D_4 roots, and the 600-cell has 120 vertices forming the H_4 root system. A finite set $X \subset S^{d-1}$ is a spherical t -design if

$$\frac{1}{|X|} \sum_{x \in X} p(x) = \frac{1}{\text{vol } S^{d-1}} \int_{S^{d-1}} p(x) \, d\Omega \quad (3.1)$$

for every polynomial p of total degree at most t (Delsarte et al., 1977). Exact coordinate checks in the corpus give

$$X_{24} \text{ is a spherical 5-design,} \quad X_{600} \text{ is a spherical 11-design.} \quad (3.2)$$

If harmonics of degree N are sampled on such a set, their Gram products have degree at most $2N$. Exact orthogonality therefore holds when $2N \leq t$. For Fock's hydrogenic momentum-space harmonics, $N = n - 1$, so the sampling limits are

$$t = 5 : \quad n \leq 3, \quad (3.3)$$

$$t = 11 : \quad n \leq 6. \quad (3.4)$$

ESTABLISHED RESULT 3.1 — WHAT THE CUTOFFS MEAN

The observed finite sampling cutoffs follow from spherical-design strength. This explains why the discrete Gram matrix is exact through those degrees and fails beyond them. It is a cubature theorem, not a derivation of hydrogen or of the Coulomb interaction.

The physical application and its circularity are treated in Volume IV. Here the important result is methodological: the loss of exactness at $n = 4$ or $n = 7$ is the predictable resolution limit of the corresponding finite design.

Admissible reduction and the readable algebra

4.1 The operator-algebraic construction

Let $\mathcal{A}$ be a unital C^* -algebra of carrier observables and $\mathcal{B} \subseteq \mathcal{A}$ a unital subalgebra. The corpus calls a reduction $C : \mathcal{A} \rightarrow \mathcal{B}$ admissible when it is unital, idempotent, completely positive, bimodular over $\mathcal{B}$, and covariant under the declared carrier symmetries. Define the minimal decay generator

$$K_0 = I - C. \quad (4.1)$$

Then

$$\ker K_0 = \text{ran } C = \mathcal{B}. \quad (4.2)$$

Equation (4.2) is exact once C is specified. Calling $\mathcal{B}$ the physical readable algebra is the framework's interpretation.

A norm-one projection from a C^* -algebra onto a C^* -subalgebra has the bimodule property of a conditional expectation, under the hypotheses of Tomiyama's theorem (Tomiyama, 1957). For a finite group G acting by automorphisms, the canonical fixed-point expectation is

$$C_G(a) = \frac{1}{|G|} \sum_{g \in G} g \cdot a, \quad (4.3)$$

whose image is $\mathcal{A}^G$.

ESTABLISHED RESULT 4.1 — FINITE-GROUP AVERAGING

Equation (4.3) is unital, positive, idempotent, covariant, and maps onto the fixed algebra. Uniqueness requires the action, target algebra, and covariance/conditional-expectation class to be fixed; covariance in the abstract does not select a group by itself.

4.2 Minimal and graded decay

Since $C^2 = C$, the minimal generator $K_0 = I - C$ is also idempotent. Its spectrum lies in $\{0, 1\}$. That is adequate for separating readable from suppressed sectors, but too coarse for a nontrivial hierarchy of masses or decay rates. A general positive generator K may satisfy

$$K\mathbf{1} = 0, \quad \ker K = \mathcal{B}, \quad (4.4)$$

while carrying more than two eigenvalues. The corpus correctly distinguishes these tasks. The later closure analysis locates the mass-squared operator in the boundary 15 and reduces numerical family splitting to a twelve-component stationary selector. Determining that stationary spectrum, together with its absolute scale, remains the localized calculation beyond the minimal projector K_0 .

4.3 The rank-three candidates

Up to the relevant Weyl conjugacy, the inherited finite enumeration identifies three rank-three root-subsystem types and their rank-five orthogonal complements:

TABLE 4.1. Rank-three candidates in the corpus enumeration.

Decayed subsystem	Orthogonal complement	Root count of complement
$A_1 \oplus A_1 \oplus A_1$	$D_4 \oplus A_1$	26
$A_2 \oplus A_1$	A_5	30
A_3	D_5	40

The corpus selects A_3 : the first candidate lacks an A_2 -type triangular root history, while the second is rejected as reducible. The first exclusion is meaningful within the adopted history rule. The second requires more care: a reducible root system does not, by itself, prove that a physical reduction must split into two independent generators. That is a model assumption about how algebraic factors map to dynamics.

CONDITIONAL SELECTION 4.1 — A_3 DECAY AND D_5 SURVIVOR

If the physical reduction is a single irreducible reflection-generated decay, if admissible decay must contain the selected triangular history, and if the rank bookkeeping of the geometry block is accepted, the surviving root algebra is $D_5 \cong \mathfrak{so}(10)$. The root-complement calculation is exact; the physical selection

premises remain conditional.

4.4 The $A_3 \times D_5$ branching

At the level of complex representations, the E_8 adjoint admits the dimensionally consistent branching

$$\mathbf{248} = (\mathbf{15}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{45}) \oplus (\mathbf{6}, \mathbf{10}) \oplus (\mathbf{4}, \mathbf{16}) \oplus (\overline{\mathbf{4}}, \overline{\mathbf{16}}). \quad (4.5)$$

The $\mathbf{16}$ of $SO(10)$ is the standard representation accommodating one Standard Model family together with a right-handed neutrino (Fritzsch and Minkowski, 1975). Equation (4.5) explains why matter-like spinor sectors appear in the finite carrier bookkeeping.

EXPOSURE 4.1 — REPRESENTATION CONTENT IS NOT YET MATTER

A root decomposition does not supply a Lorentz-covariant quantum field, spin-statistics, chirality, a kinetic term, a gauge coupling, or a mass matrix. The claimed split of a $\mathbf{4}$ index into one invariant direction plus three families is a proposed physical reading, not a standard consequence of Equation (4.5).

Direct attempts to embed Lorentz and Standard Model gauge structure in a real or complex form of E_8 face representation-theoretic no-go results. Distler and Garibaldi state precise subgroup and chirality hypotheses for the class they exclude (Distler and Garibaldi, 2010). The corpus argues that its finite decay group is not the Lorentz subgroup assumed there. That observation is not yet an evasion: a complete model must first provide its own Lorentz action and chiral field content, after which the theorem's applicability can be assessed exactly.

Connections, holonomy, and the co-distinction proposal

5.1 Established geometric structure

On a principal bundle, a connection one-form A specifies parallel transport and has curvature

$$F = dA + A \wedge A. \quad (5.1)$$

For a small loop $\gamma = \partial\Sigma$, holonomy differs from the identity at leading order by the curvature flux through Σ . In a gauge theory, A and F take values in a Lie algebra; Yang and Mills introduced the non-Abelian field strength and its gauge dynamics in physics (Yang and Mills, 1954). Geometric phase supplies another physical manifestation of holonomy (Berry, 1984; Aharonov and Anandan, 1987).

The corpus's useful conceptual statement is that ordered transport can fail to close even when the projected endpoint returns. Equation (5.1) already formalizes this for connections. The new proposal is to treat distinguishable actions themselves as the transported quantity and to select among their cycles by a least-cost rule.

5.2 The Shields field as a proposed extension

Let a denote a point in a parameter or action space, and let $\mathcal{A}^\alpha_i(a)$ be a connection-like object mapping changes in that space to a co-distinction displacement. The proposed loop response is

$$\Delta X^\alpha[\gamma] = \oint_\gamma \mathcal{A}^\alpha_i(a) da^i. \quad (5.2)$$

Nonzero loop response is called a radiance in the corpus; variation across paths is associated with scattering or curvature. These names are not yet physical fields. A field theory requires a base manifold, fibre, transformation law, curvature, source, action, dimensions, and coupling to instruments.

RESEARCH TARGET 5.1 — LEAST-COST HOLONOMY

Define a gauge-invariant functional on the holonomy classes of Equation (5.2), prove existence and stability of its minimisers, and derive an observable coupling that distinguishes the proposal from ordinary geometric phase or radiative transport.

Interrogation can change the holonomy class when the measurement coupling becomes part of the loop. That observation is sound but general: a measurement intervention changes the dynamics unless it commutes with them. The new content must lie in a quantitative law for the change.

The gravitational square

6.1 MacDowell–Mansouri gravity

MacDowell and Mansouri formulated gravity using a de Sitter-type gauge connection and a projection to the Lorentz subalgebra (MacDowell and Mansouri, 1977). In schematic first-order notation, decompose the curvature as

$$F^{ab} = R^{ab} - \ell^{-2} e^a \wedge e^b, \quad F^{a4} = \ell^{-1} T^a, \quad (6.1)$$

where e^a is a coframe, R^{ab} the Lorentz curvature, T^a torsion, and ℓ a length scale. The projected quadratic action contains

$$\int \epsilon_{abcd} \left(R^{ab} \wedge R^{cd} - 2\ell^{-2} R^{ab} \wedge e^c \wedge e^d + \ell^{-4} e^a \wedge e^b \wedge e^c \wedge e^d \right). \quad (6.2)$$

The first term is topological in four dimensions under the usual conditions, the mixed term gives Einstein–Cartan/Einstein–Hilbert dynamics, and the final term is cosmological.

6.2 The corpus coefficient proposal

The corpus forms an abstract defect from a curvature-like object and a realised-area object,

$$\mathcal{D} = u \mathcal{F} - r \mathcal{E}, \quad \langle \mathcal{D}, \mathcal{D} \rangle = u^2 \langle \mathcal{F}, \mathcal{F} \rangle - 2ur \langle \mathcal{F}, \mathcal{E} \rangle + r^2 \langle \mathcal{E}, \mathcal{E} \rangle. \quad (6.3)$$

The coefficient pattern matches the three-term structure of Equation (6.2). Moreover,

$$\frac{r}{u} = \frac{1}{\sqrt{5}}, \quad 3\frac{r}{u} = \frac{3}{\sqrt{5}}. \quad (6.4)$$

These are exact internal ratios.

CONDITIONAL IDENTIFICATION 6.1 — FROM THE CARRIER SQUARE TO GRAVITY

Equation (6.3) becomes a MacDowell–Mansouri-type gravitational action only if $\mathcal{F}$, $\mathcal{E}$, the invariant pairing, form degrees, gauge group, symmetry reduction, dimensional scale ℓ , and boundary terms are specified so that the equality holds

as an action, not merely as a three-term polynomial pattern.

The corpus further proposes that its cluster projection supplies the symmetry-reducing projection in MacDowell–Mansouri gravity. This is a valuable concrete bridge to test: both maps must be written in a common representation and shown equal or naturally equivalent. At present the identification is proposed, not proved.

6.3 Variation and torsion

For the appropriate first-order gravitational action, independent variation with respect to coframe and connection yields an Einstein equation with cosmological term and a torsion equation. In vacuum or for matter without intrinsic spin, the torsion equation commonly gives $T^a = 0$; spinful matter can source torsion in Einstein–Cartan theory (Hehl et al., 1976). The inherited blanket statement “torsion vanishes rather than being imposed” is therefore too broad.

EXPOSURE 6.1 — FIELD EQUATIONS ARE ONLY AS DERIVED AS THE ACTION MAP

Standard variation of a standard first-order action is borrowed mathematics. It does not validate the corpus action until the map in Conditional Identification 6.1 is complete. Couplings to the proposed matter sector must also be included before the torsion and stress-energy equations can be stated.

Least boundary and the compression field

7.1 The single-cell theorem

Let $\Omega \subset \mathbb{R}^3$ be a bounded region of fixed volume V , let $\eta > 0$ be an isotropic boundary distinction density, and define the instantaneous informational boundary cost

$$\dot{\mathcal{J}}_\partial[\Omega] = W_B \eta |\partial\Omega|. \quad (7.1)$$

The Euclidean isoperimetric inequality gives

$$|\partial\Omega| \geq (36\pi)^{1/3} V^{2/3}, \quad (7.2)$$

with equality for a ball under the regularity hypotheses (Osseman, 1978). Multiplication by the positive constant $W_B \eta$ preserves the minimiser.

INTERNAL RESULT 7.1 — FINITE-READOUT CELL THEOREM

Given Equation (7.1), Euclidean geometry, isotropic η , and fixed volume, a ball uniquely minimises the boundary cost. If a readout capacity C imposes $W_B \eta |\partial\Omega| \leq C$, the saturated ball has

$$R = \sqrt{\frac{C}{4\pi W_B \eta}}. \quad (7.3)$$

The geometry is proved; interpreting η , C , and W_B as physical information variables is conditional.

For a prescribed volume history $V(t)$, applying Equation (7.2) at each time and integrating proves that the pointwise least-boundary history minimizes

$$\mathcal{J}_\partial = \int W_B \eta |\partial\Omega(t)| dt. \quad (7.4)$$

This is a valid conditional variational theorem. It does not derive a physical Lagrangian unless a physical action is shown to represent the same conversion count.

7.2 From isolated cells to saturation

Equal spheres cannot tile Euclidean three-space; their densest packing fraction is $\pi/\sqrt{18}$, as established by the Kepler-conjecture proof (Hales, 2005). A space-filling equal-volume partition is therefore a different variational problem from the isolated isoperimetric problem. Shared interfaces, Plateau conditions, topology, and periodicity enter.

Weaire and Phelan found a partition with lower interface area than Kelvin's candidate (Weaire and Phelan, 1994). It remains a best-known structure, not a proved global optimum. The inherited text is correct to flag this but too strong when it says the saturated shape is proved.

CONDITIONAL IDENTIFICATION 7.1 — SATURATED LEAST-INTERFACE FIELD

If persistent resolution requires a gap-free equal-volume partition and if total shared interface is the complete cost, saturation selects a minimiser of the three-dimensional Kelvin problem. The minimiser's existence class and local Plateau laws are standard; its global identity is not known. Weaire–Phelan is a candidate, not a theorem.

For anisotropic boundary density $\eta(\hat{n})$, the minimiser is generally a Wulff shape rather than a sphere. This provides a natural falsification boundary: anisotropic fields should deform the cells in a calculable direction.

7.3 The Bubble Field as a compression map

The later Bubble Field is best understood as a finite-dimensional compression, not a new fundamental field by declaration. Its local state is

$$x = (\alpha_M, \alpha_L, \nu, \theta), \quad (7.5)$$

and the compressed variables are

$$C(x) = (A, B, \Theta) = (-\alpha_M, \alpha_L + \nu/3, \theta). \quad (7.6)$$

The Jacobian has rank three and kernel

$$\ker DC = \text{span}\{(0, -1, 3, 0)\}. \quad (7.7)$$

Thus the metric sector cannot separately recover α_L and ν . Any claimed output depending on that split needs an additional measurement or constitutive section.

INTERNAL RESULT 7.2 — EXACT FIELD SUFFICIENCY

A quantity $Q(\alpha_M, \alpha_L, \nu, \theta)$ is recoverable from the Bubble Field exactly when it is invariant along Equation (7.7), equivalently when $Q = \tilde{Q}(A, B, \Theta)$. Metric, curvature, geodesic, clock, and lensing formulas written only in A and B satisfy this criterion by construction.

The proposed least-response constitutive section minimizes a Euclidean quadratic cost under the mixed constraint $r\alpha_L + u\alpha_M = \chi$, giving the ratio $\alpha_L/\alpha_M = r/u = 1/\sqrt{5}$. This is an exact Lagrange-multiplier result for the chosen cost; the cost itself is a testable constitutive assumption.

7.4 Spherical metric reconstruction

In isotropic coordinates the field uses

$$ds^2 = -e^{2A(\rho)}c^2dt^2 + e^{2B(\rho)}(d\rho^2 + \rho^2d\Omega^2). \quad (7.8)$$

The exterior Schwarzschild solution is represented exactly by

$$A = \ln \frac{1-x}{1+x}, \quad B = 2 \ln(1+x), \quad x = \frac{GM}{2\rho c^2}. \quad (7.9)$$

This is a correct coordinate re-expression of established general relativity, originating with Einstein's field theory (Einstein, 1916). It demonstrates that the compression variables can encode a known spherical metric. It does not derive that metric from the microscopic cell variables.

EXPOSURE 7.1 — RECONSTRUCTION VERSUS DYNAMICS

Equations (7.8)–(7.9) are top-down reconstructions. A bottom-up field theory still requires equations for $\alpha_M, \alpha_L, \nu, \theta$, a source action, boundary conditions, and a derivation of the constitutive section.

The source sector and present frontier

8.1 Local and bilocal fields

A local scalar action such as

$$S_\phi = \int \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right] d^4x \quad (8.1)$$

has a local stress-energy tensor obtained by metric variation. A bilocal object $\Phi(x, y)$ does not automatically define a local source at x . One must specify a kernel, diagonal restriction, integration over y , or localization map and show covariant conservation of the resulting tensor.

The corpus’s source-sector papers correctly isolate this issue. Orbit data determine $\mu = GM$, not G and M separately. Likewise, a geometric reconstruction can identify a metric potential without determining the microscopic source that produced it.

ESTABLISHED IDENTIFIABILITY FACT 8.1

In Newtonian two-body orbital dynamics, trajectories constrain the gravitational parameter $\mu = GM$. Separating G and M requires an independent mass or coupling measurement. The same product degeneracy must be respected by any proposed source reconstruction.

8.2 What is exact, what is proposed

TABLE 8.1. Carrier and field ledger.

Statement	Grade	Boundary of the result
Uniqueness of rank-eight positive-definite even unimodular lattice	established	conditional carrier selection because the hypotheses are not yet derived
240-root coordinates and root reflections	established	physical step interpretation is additional

Statement	Grade	Boundary of the result
Triangular root closure and areas	exact	physical history/cost is conditional
Golden Coxeter/ H_4 geometry	established plus certified	no physical coupling follows from geometry alone
Spherical-design sampling limits	exact	cubature, not atomic dynamics
Finite-group algebra average	fixed- established	group and physical algebra must be specified
A_3 complement D_5	exact finite algebra once selected	selection and family reading are conditional
Co-distinction holonomy field	conjecture	requires bundle, action, coupling, and measurement
Three-term gravitational square	exact polynomial pattern	equality to a gravitational action is conditional
Single-cell least-boundary shape	internal theorem from standard geometry	physical information cost is a postulate
Saturated foam identity	open	Weaire–Phelan is best-known, not proven optimal
Bubble compression rank and kernel	exact	dynamics and constitutive section remain conditional

8.3 Forward obligations

The carrier programme has four load-bearing tasks:

1. derive the lattice hypotheses and rank from the traversal grammar;
2. construct the physical reduction and show why it is the A_3 average;
3. write a dimensionally complete action linking the carrier square to gauge, matter, and gravitational fields;
4. derive an observable source and coupling rather than reconstructing known metrics from chosen potentials.

Volume III now turns from the carrier to what can be observed from it: causal order, traversal depth, field assembly, measurement architecture, and the losses introduced by each readout.

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